Cantera vs NN | Test IC 64 | NRMSE (util.py)

T0 = 1045.3 K P0 = 4.301 atm phi0 = 0.595 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**10k**	0.284	0.791	1.02	1.04	0.509	11.4	0.523	1.02	1.01	0	Complete
2 P-only DeepONet10	**10k**	0.542	1.37	1	1	0.733	1	1	0.999	1	0	Complete
2 P-only DeepONet10	**long**	0.336	0.848	1.02	1.81	0.455	0.553	0.622	1.05	1.43	0	Complete
3 ANN10 + P-only DeepONet10	**10k**	0.542	1.37	1	1	0.733	1	1	0.999	1	0	Complete
3 ANN10 + P-only DeepONet10	**long**	0.294	0.742	0.98	1.61	0.399	0.504	0.545	1.18	1.79	0	Complete
4 T/P DeepONet10	**10k**	0.346	0.949	6.56	1.71	0.539	3.79	0.654	1.21	0.992	0	Complete
4 T/P DeepONet10	**long**	0.557	0.963	54.9	0.698	0.903	39.2	0.734	1.01	1	0	Complete
5 ANN10 + T/P DeepONet10	**10k**	0.331	0.919	4.92	1.11	0.528	4.59	0.625	1.09	0.99	0	Complete
5 ANN10 + T/P DeepONet10	**long**	0.406	0.874	27.7	0.985	0.74	28.8	0.611	0.916	0.996	0	Complete
6 T/P/ϕ DeepONet10	**10k**	0.542	1.37	1	1	0.733	1	1	0.999	1	0	Complete
6 T/P/ϕ DeepONet10	**long**	0.542	1.37	1	1	0.733	1	1	0.999	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**10k**	0.542	1.37	1	1	0.733	1	1	0.998	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**long**	0.312	0.881	59.7	0.777	0.532	11.6	0.548	2.86	0.99	0	Complete

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
ANN stays at 10k in every mixed team. Long DeepONets stopped at their wall-time caps; no verified loss plateau.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.